HSC Mathematics Advanced — Practice Paper

Topic: Sequences and Series (Arithmetic & Geometric) — Student Version

Questions adapted from NSW school Trial HSC Examinations 2025–2026 (MA12-4) • 79 questions, 208 marks

General instructions
• Reading time – 5 minutes • Working time – 2 hours 30 minutes
• Write using black pen • NESA-approved calculators may be used
• For Questions 1–21 (multiple choice), circle the correct answer
• For Questions 22–79, write your response in the space provided and show relevant mathematical reasoning and/or calculations

Difficulty labels EASY MODERATE DIFFICULT

Question 1 (1 mark) EASY

Adapted from Blacktown Boys High School 2025 Trial HSC

Which of the following is an arithmetic sequence?

(A) $6,\ 24,\ 96,\ \ldots$ (B) $9,\ 14,\ 20,\ \ldots$ (C) $27,\ 34,\ 41,\ \ldots$ (D) $48,\ -24,\ 12,\ \ldots$

Question 2 (1 mark) EASY

Adapted from Blacktown Boys High School 2026 Trial HSC

The first three terms of an arithmetic sequence are 5, 9 and 13. What is the 18th term of the sequence?

(A) 81 (B) 77 (C) 73 (D) 69

Question 3 (1 mark) EASY

Adapted from North Sydney Boys High School 2026 Trial HSC

What is the 20th term of the arithmetic sequence $-7,\ -3,\ 1,\ 5,\ \ldots$?

(A) 65 (B) 69 (C) 73 (D) 77

Question 4 (1 mark) EASY

Adapted from Manly Selective Campus 2026 Trial HSC

The numbers $5,\ x,\ 13,\ 17,\ \ldots$ form an arithmetic sequence. What is the sum of the first four terms?

(A) 35 (B) 39 (C) 42 (D) 44

Question 5 (1 mark) EASY

Adapted from Fort Street High School 2026 Trial HSC

In an arithmetic series, the first term is -2 and the third term is 8. What is the 10th term?

(A) 38 (B) 41 (C) 43 (D) 45

Question 6 (1 mark) EASY

Adapted from Caringbah High School 2026 Trial HSC

In an arithmetic series, the first term is -2 and the third term is 8. What is the sum of the first 15 terms?

(A) 395 (B) 450 (C) 495 (D) 540

Question 7 (1 mark) EASY

Adapted from Hunters Hill High School 2026 Trial HSC

The first three terms of an arithmetic sequence are $1,\ 3,\ 5,\ \ldots$. Which of the following gives the sum of the first n terms, S_n?

(A) $S_n = n^2$ (B) $S_n = 2n$ (C) $S_n = 2n - 1$ (D) $S_n = 2n + 1$

Question 8 (1 mark) EASY

Adapted from Hunters Hill High School 2025 Trial HSC

Find the limiting sum of the series $2\pi + \pi + \frac{\pi}{2} + \cdots$

(A) $\frac{\pi}{2}$ (B) π (C) 2π (D) 4π

Question 9 (1 mark) **EASY**

Adapted from St George Girls High School 2025 Trial HSC

Consider the series $1+1+1+1+\cdots$. Which statement could NOT be made about the series?

(A) It is an arithmetic progression with common difference 0

(B) It is a geometric progression with common ratio 1

(C) It has a limiting sum of $2n$, where n is the number of terms

(D) Its terms satisfy $T_n = T_{n-1}$, where $T_1 = 1$

Question 10 (1 mark) **EASY**

Adapted from Blacktown Boys High School 2025 Trial HSC

Find the 100th term of the sequence $10,\ 6\frac{1}{2},\ 3,\ \ldots$

(A) $356\frac{1}{2}$ (B) $-336\frac{1}{2}$ (C) -333 (D) 353

Question 11 (1 mark) **MODERATE**

Adapted from Killara High School 2025 Trial HSC

Consider the geometric sequence $81,\ a,\ b,\ 3$. Find the value of $b-a$.

(A) -18 (B) 18 (C) -26 (D) 26

Question 12 (1 mark) **MODERATE**

Adapted from Hurlstone Agricultural High School 2026 Trial HSC

If $x^2,\ 4x+3,\ 25$ are in a geometric sequence, what is the value of x?

(A) $-\frac{1}{3}$ (B) -3 (C) $\frac{1}{3}$ or -3 (D) $-\frac{1}{3}$ or 3

Question 13 (1 mark) **MODERATE**

Adapted from Killara High School 2026 Trial HSC

A geometric series is given by $4+2x+x^2+\frac{x^3}{2}+\cdots$. If the limiting sum of the series is 10, what is the value of x?

(A) $\frac{1}{2}$ (B) $\frac{3}{5}$ (C) $\frac{4}{5}$ (D) $\frac{6}{5}$

Question 14 (1 mark) **MODERATE**

Adapted from Blacktown Boys High School 2025 Trial HSC

Which expression is a term of the geometric series $2x-4x^2+8x^3-16x^4+\cdots$?

(A) $-2048x^{11}$ (B) $2048x^{11}$ (C) $-2048x^{10}$ (D) $2048x^{10}$

Question 15 (1 mark) **MODERATE**

Adapted from North Sydney Boys High School 2025 Trial HSC

For an arithmetic sequence, $T_4+T_5=24$ and $S_6=48$. What is the common difference d?

(A) 1 (B) 2 (C) 4 (D) 8

Question 16 (1 mark) **MODERATE**

Adapted from Penrith High School 2025 Trial HSC

The formula for the nth term of a series is $T_n=400n-2^n$. What is the sum of the first 10 terms?

(A) 2 976 (B) 407 154 (C) $-135\,718$ (D) 19 954

Question 17 (1 mark) **MODERATE**

Adapted from Girraween High School 2025 Trial HSC

The first term of a geometric series is 4 and the third term is 9. The common ratio is negative. The 40th term of the series is

(A) $4\left(\frac{3}{2}\right)^{39}$ (B) $-4\left(\frac{3}{2}\right)^{39}$ (C) $4\left(\frac{3}{2}\right)^{40}$ (D) $-4\left(\frac{3}{2}\right)^{40}$

Question 18 (1 mark) MODERATE

Adapted from Manly Selective Campus 2025 Trial HSC

Let $0 < q < p$. If p and q are the 1st and 2nd terms of a geometric progression, what is the limiting sum?

(A) $\frac{p}{1-q}$ (B) $\frac{pq}{q-p}$ (C) $\frac{p^2}{p+q}$ (D) $\frac{p^2}{p-q}$

Question 19 (1 mark) DIFFICULT

Adapted from Manly Selective Campus 2025 Trial HSC

In a geometric sequence $T_3 = -45$, $S_5 = -305$ and $S_6 = 910$. What is the 9th term of the sequence?

(A) $-32\,805$ (B) $-10\,935$ (C) $10\,935$ (D) $32\,805$

Question 20 (1 mark) DIFFICULT

Adapted from Fort Street High School 2026 Trial HSC

The limiting sum of a geometric series is $\frac{10}{9}$ and the sum of its first three terms is $\frac{26}{25}$. What is the common ratio of this series?

(A) 1.39 (B) 1.25 (C) 0.40 (D) 0.25

Question 21 (1 mark) DIFFICULT

Adapted from Abbotsleigh 2025 Trial HSC

What is the limiting sum of the geometric series $\log_e 2 - (\log_e 2)^2 + (\log_e 2)^3 - (\log_e 2)^4 + \ldots$?

(A) $\log_{\frac{e}{2}} 2$ (B) $\log_{\frac{2}{e}} 2$ (C) $\log_{2+e} 2$ (D) $\log_{2e} 2$

Question 22 (2 marks) EASY

Adapted from Barker College 2025 Trial HSC

Find the limiting sum of the series $96 + 72 + 54 + \cdots$ **2**

Question 23 (3 marks) EASY

Adapted from Blacktown Boys High School 2025 Trial HSC

Consider the geometric series $18+6+2+\cdots$

(a) Find the common ratio. **1**

(b) Find the 8th term. **1**

(c) Find the limiting sum of this series. **1**

Question 24 (2 marks) EASY

Adapted from Blacktown Boys High School 2025 Trial HSC

In an arithmetic sequence, the sixth term is 37 and the nineteenth term is 193. Find the value of the first term, a, and the common difference, d. **2**

Question 25 (2 marks) **EASY**

Adapted from Blacktown Boys High School 2026 Trial HSC

Find the sum of the terms in the arithmetic series $2026 + 2011 + 1996 + \cdots + 271$. **2**

Question 26 (2 marks) **EASY**

Adapted from Blacktown Boys High School 2026 Trial HSC

In a geometric sequence, the 10th term is 768 and the 13th term is 6144. Find the first term and the common ratio. **2**

Question 27 (3 marks) **EASY**

Adapted from Baulkham Hills High School 2026 Trial HSC

An arithmetic sequence from start to end is $51,\ 44,\ 37,\ \ldots,\ -47$.

(a) Write a simplified expression for the nth term of this sequence. **1**

(b) How many terms are there in the sequence? **1**

(c) Find the sum of this series. **1**

Question 28 (2 marks) **EASY**

Adapted from Hornsby Girls High School 2025 Trial HSC

The first three terms of an arithmetic sequence are $x+6,\ 2x-5$ and $4x-11$. Find the value of x. **2**

Question 29 (4 marks) EASY

Adapted from Hunters Hill High School 2025 Trial HSC

The tenth term of an arithmetic series is 29 and the fifteenth term is 44.

(a) Find the value of the common difference and the value of the first term. **2**

(b) Find the sum of the first 75 terms. **2**

Question 30 (2 marks) EASY

Adapted from Hunters Hill High School 2026 Trial HSC

The numbers $1280,\ p,\ q,\ 20$ form a geometric sequence. Find the values of p and q. **2**

Question 31 (3 marks) EASY

Adapted from Killara High School 2025 Trial HSC

Find the sum of the terms in the geometric series $2+6+18+54+\cdots+118\,098$. **3**

Question 32 (2 marks) EASY

Adapted from Knox Grammar School 2026 Trial HSC

Find the sum of the arithmetic series $180+165+150+\cdots-15-30$. **2**

Question 33 (2 marks) EASY

Adapted from Manly Selective Campus 2026 Trial HSC

In an arithmetic sequence, the 6th term is 36 and the 9th term is 28. Find the first term. **2**

Question 34 (2 marks) **EASY**

Adapted from Manly Selective Campus 2025 Trial HSC

A sequence is defined recursively by $T_1 = 256$ and $T_n = \frac{T_{n-1}}{4}$.

(a) List the first four terms of the sequence. **1**

(b) Find the limiting sum of the series formed by the sequence, if it exists. If it does not exist, explain why. **1**

Question 35 (3 marks) **EASY**

Adapted from Penrith High School 2025 Trial HSC

Find the sum of the terms in the arithmetic series $71 + 64 + 57 + \cdots + (-48)$. **3**

Question 36 (2 marks) **EASY**

Adapted from Roseville College 2026 Trial HSC

An arithmetic series has a common difference of 6 and its 20th term is 130. Find the sum of the first twenty terms of the series. **2**

Question 37 (3 marks) **EASY**

Adapted from St George Girls High School 2025 Trial HSC

The nth term of an arithmetic sequence is given by $T_n = 5n - 3$.

(a) Write down the first 3 terms of the sequence. **1**

(b) Find the sum of the first 50 terms. **2**

Question 38 (2 marks) **EASY**

Adapted from St George Girls High School 2025 Trial HSC

Determine the last term under 100 in the sequence $-11,\ -4,\ 3,\ 10,\ 17,\ \ldots$ **2**

Question 39 (2 marks) **EASY**

Adapted from Sydney Girls High School 2025 Trial HSC

Consider the geometric series $7+14e^{-2}+28e^{-4}+56e^{-6}+\cdots$. Justify whether this series has a limiting sum or not. **2**

Question 40 (2 marks) **EASY**

Adapted from Sydney Girls High School 2026 Trial HSC

The first term of an arithmetic series is -7 and the sum of the first 20 terms is 430. Find the common difference. **2**

Question 41 (2 marks) **EASY**

Adapted from Sydney Grammar School 2025 Trial HSC

Find the value of the arithmetic series $1+5+9+\cdots+2025$. **2**

Question 42 (2 marks) **EASY**

Adapted from Sydney Technical High School 2026 Trial HSC

Find the first term of the series $-12-8-4+\cdots$ that is greater than 150. **2**

Question 43 (2 marks) **EASY**

Adapted from North Sydney Boys High School 2026 Trial HSC

Find the sum of the arithmetic series $2+5+8+\cdots+53$. **2**

Question 44 (3 marks) **EASY**

Adapted from Sydney Girls High School 2025 Trial HSC

Consider the arithmetic sequence $-7,\ 2,\ 11,\ 20,\ \ldots,\ 713$.

(a) How many terms are in the sequence? **1**

(b) Find the sum of all the terms. **2**

Question 45 (3 marks) **EASY**

Adapted from St George Girls High School 2025 Trial HSC

Find the sum of the geometric series $2+6+18+\cdots+39\,366$. **3**

Question 46 (4 marks) MODERATE

Adapted from Barker College 2025 Trial HSC

An arithmetic series is given by $118+105+92+79+\cdots$

(a) Find the 10th term of the series. **1**

(b) Find the minimum number of terms needed for the sum of the series to be negative. **3**

Question 47 (2 marks) MODERATE

Adapted from Blacktown Boys High School 2026 Trial HSC

Evaluate the limiting sum of the series $\frac{5}{6}+\frac{5}{27}+\frac{10}{243}+\ldots$ **2**

Question 48 (5 marks) MODERATE

Adapted from Caringbah High School 2025 / Hills Grammar 2025 Trial HSC

Finn accidentally puts a hole in the bottom of his rainwater tank, which contains 10 200 litres of water. In the first hour 200 litres leak out, and in the second hour a further 250 litres leak out. Finn wonders whether the amount leaking each hour forms an arithmetic or a geometric sequence.

(a) If the amount of water that leaks out each hour forms an arithmetic sequence, how many hours will it take until the tank is empty? **3**

(b) If the amount of water that leaks out each hour forms a geometric sequence, show that eight hours after the leak begins, over 60% of the water remains in the tank. **2**

Question 49 (3 marks) MODERATE

Adapted from Hornsby Girls High School 2025 Trial HSC

The sum of the first n terms of an arithmetic sequence is $S_n = 3n^2 - 4n$, where $n > 1$. Find the nth term, T_n, and state the common difference. **3**

Question 50 (5 marks) MODERATE

Adapted from Hornsby Girls High School 2026 Trial HSC

A musical instrument has many strings. The difference between the lengths of adjacent strings is constant, so the lengths form an arithmetic sequence. The shortest string is 30 cm and the longest is 50 cm. The sum of the lengths of all the strings is 1240 cm.

(a) Find the number of strings. **2**

(b) Find the difference in length between adjacent strings. **2**

(c) What is the length of the tenth string? **1**

Question 51 (3 marks) MODERATE

Adapted from Hurlstone Agricultural High School 2026 Trial HSC

In an arithmetic series, the 1st term is -12, the 17th term is 12 and the last term is 45. Find the sum of all the terms in the series. **3**

Question 52 (5 marks) MODERATE

Adapted from James Ruse Agricultural High School 2025 Trial HSC

A venetian blind consists of twenty slats, each 2.5 mm thick. When the blind is down, the gap between the top slat and the top of the blind is 27 mm and the gap between adjacent slats is also 27 mm. When the blind is up, all the slats are stacked at the top with no gaps.

(a) Show that when the blind is raised, the bottom slat rises 540 mm. **2**

(b) How far does the 19th slat rise? **1**

(c) Explain briefly why the distances the slats rise form an arithmetic sequence. **1**

(d) Find the sum of all the distances that the slats rise when the blind is raised. **1**

Question 53 (3 marks) MODERATE

Adapted from Killara High School 2026 Trial HSC

In an arithmetic sequence, the sum of the first six terms is 78 and the sum of the first ten terms is 210. Find the first term and the common difference. **3**

Question 54 (4 marks) MODERATE

Adapted from James Ruse Agricultural High School 2026 Trial HSC

A wildlife camera records video clips over consecutive nights. The number of clips recorded each night forms an arithmetic sequence. Over the first 10 nights the camera records a total of 730 clips. On the tenth night it records 100 clips.

(a) Show that the number of clips recorded on the first night is 46. **1**

(b) Find the common difference of the sequence. **1**

(c) Assuming the same pattern continues, find the first night by which the total number of clips recorded exceeds 1200. **2**

Question 55 (2 marks) MODERATE

Adapted from Merewether High School 2025 Trial HSC

The sum of the first 10 terms of an arithmetic sequence is 160. The sum of the first 20 terms is 520. What is the first term a of the sequence? **2**

Question 56 (4 marks) MODERATE

Adapted from Normanhurst Boys High School 2026 Trial HSC

The sales team at Frontier phone company sells 12 000 phones in the first month of operation. They increase their sales by 800 phones each month on the preceding month's sales.

(a) Find the number of phones sold in the last month of the second year of operation. **1**

(b) Sampson, another phone company, commenced business at exactly the same time. Their sales team sells 5000 phones in their first month and increases sales by 1500 each month on the preceding month's sales. After how many months will Sampson's total sales exceed Frontier's total sales? **3**

Question 57 (4 marks) MODERATE

Adapted from North Sydney Girls High School 2025 Trial HSC

For a geometric series the third term is 20 and the sixth term is -160.

(a) Find the common ratio and the first term. **2**

(b) Find the sum of the terms from T_{10} to T_{20} inclusive. **2**

Question 58 (5 marks) MODERATE

Adapted from North Sydney Boys High School 2025 Trial HSC

Consider the series $8+12x+18x^2+27x^3+\cdots$

(a) Show that the terms in this series form a geometric progression. **1**

(b) For what value(s) of x will this series have a limiting sum? **2**

(c) Find the limiting sum if $x=\frac{1}{4}$. **2**

Question 59 (5 marks) MODERATE

Adapted from North Sydney Girls High School 2026 Trial HSC

Given the arithmetic series $27+x+y+z+11+\cdots$

(a) Find the values of x, y and z. **2**

(b) Find the number of terms if the series sums to 99. **3**

Question 60 (3 marks) MODERATE

Adapted from St George Girls High School 2025 Trial HSC

The sequence $6-x,\ x-8,\ x+4$ is geometric. Determine the possible values of the common ratio. **3**

Question 61 (2 marks) MODERATE

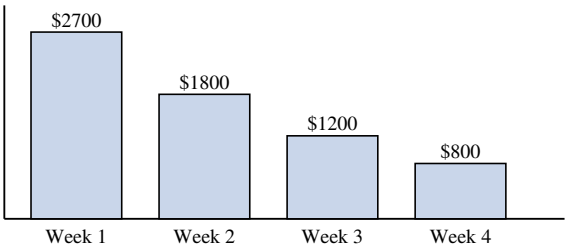
Adapted from Sydney Grammar School 2025 Trial HSC

Find the values of x such that the limiting sum $1+(1-2x)+(1-2x)^2+(1-2x)^3+\cdots$ exists. **2**

Question 62 (5 marks) **MODERATE**

Adapted from St George Girls High School 2026 / Roseville College 2026 Trial HSC

A charity records the donations received each week for the first four weeks of a fundraising drive, as shown. The director notices that the weekly donations follow a geometric sequence and assumes the pattern will continue.



(a) Find the common ratio. **1**

(b) How much would she predict to be received in Week 8 (to the nearest dollar)? **1**

(c) What total amount would be raised in the first thirteen weeks? **1**

(d) She claims the drive will never raise more than \$8100 in total. Explain how she can make this prediction and why she decides to launch a new drive after thirteen weeks. **2**

Question 63 (4 marks) MODERATE

Adapted from Sydney Grammar School 2026 Trial HSC

A scientist prepares solutions whose hydrogen ion concentrations form a geometric sequence. Let H_n be the concentration of the nth solution. The first solution has concentration $H_1 = 10^{-2}$ mol/L and each subsequent solution has half the concentration of the previous one.

(a) Find the concentration of hydrogen ions in the nth solution. **1**

(b) The acidity of a solution is its pH value, $A(H) = -\log_{10} H$. Show that the acidity of the nth solution is $A_n = 2 + (n-1)\log_{10} 2$. **2**

(c) Determine the first solution in the sequence whose acidity exceeds 7. **1**

Question 64 (3 marks) DIFFICULT

Adapted from Baulkham Hills High School 2025 Trial HSC

The common difference of an arithmetic sequence is -4. The sum of the first 30 terms is 23 940 and the sum of the last 30 terms is -1980. Find how many terms are in the sequence. **3**

Question 65 (3 marks) **DIFFICULT**

Adapted from Caringbah High School 2026 / Fort Street High School 2026 Trial HSC

Consider the function defined by the geometric series $f(x) = e^{-x} + e^{-2x} + e^{-3x} + \cdots$ for $x > 0$.

(a) Explain why the limiting sum exists for $x > 0$. **1**

(b) Show that $f(x) = \dfrac{1}{e^x - 1}$. **2**

Question 66 (5 marks) **DIFFICULT**

Adapted from Girraween High School 2025 Trial HSC

Consider the geometric series $1 - \tan^2\theta + \tan^4\theta - \cdots$

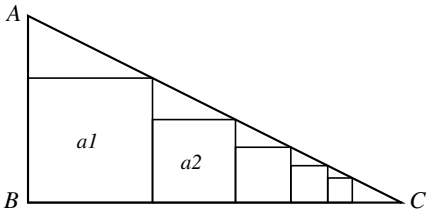
(a) When the limiting sum exists, find an expression for the limiting sum in simplest form. **3**

(b) For what values of θ in the interval $-\dfrac{\pi}{2} < \theta < \dfrac{\pi}{2}$ does the limiting sum of the series exist? **2**

Question 67 (5 marks) DIFFICULT

Adapted from Fort Street High School 2025 Trial HSC

A sequence of squares is inscribed within a right-angled triangle ABC, where $AB = a$, $BC = 2a$ and $\angle ABC = 90°$. Each square touches the hypotenuse AC, and the side lengths are $a_1,\ a_2,\ \ldots,\ a_n$.



(a) By considering similar triangles, show that $a_1 = \frac{2}{3}a$. **1**

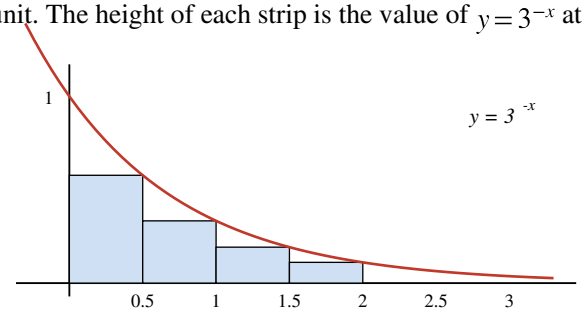
(b) Show that the areas of the squares form a geometric sequence. State the common ratio. **2**

(c) Find, in terms of a, the total sum of the areas of all the squares. **2**

Question 68 (3 marks) **DIFFICULT**

Adapted from Hornsby Girls High School 2026 Trial HSC

The diagram shows the graph of $y = 3^{-x}$ and the first four of an infinite number of rectangular strips of width $\frac{1}{2}$ unit. The height of each strip is the value of $y = 3^{-x}$ at the right-hand edge of the strip, i.e. at $x = \frac{1}{2},\ 1,\ \frac{3}{2},\ \ldots$



(a) The sum of the areas of the rectangles forms a geometric series. Show that the limiting sum of this series is $\frac{1}{2\sqrt{3}-2}$. **2**

(b) Express the limiting sum with a rational denominator. **1**

Question 69 (5 marks) **DIFFICULT**

Adapted from Hunters Hill High School 2026 / Normanhurst Boys 2025 Trial HSC

ABC is a triangle. The numbers $1+\sin A,\ \frac{1}{2},\ 1-\sin A$ are the first three terms of a geometric sequence.

(a) Find the possible values of A. **3**

(b) Show that the limiting sum of the series $(1+\sin A)+\frac{1}{2}+(1-\sin A)+\cdots$ exists. **2**

Question 70 (4 marks) **DIFFICULT**

Adapted from James Ruse Agricultural High School 2025 Trial HSC

Given the series $1\frac{1}{3}+2\frac{1}{9}+4\frac{1}{27}+8\frac{1}{81}+\ldots$, find an expression for

(a) the general term. **2**

(b) the sum of the first n terms. **2**

Question 71 (5 marks) DIFFICULT

Adapted from Knox Grammar School 2026 Trial HSC

Two arithmetic progressions, A and B, each have 100 terms denoted by a_i and b_i respectively, where $i = 1, 2, \ldots, 100$. The common difference of A is d, a positive integer. The progressions satisfy $a_1 = b_{100} = 4$ and $b_1 = a_{100}$.

(a) You are given that there is at least one value of i for which $b_i = 10 + a_i$. Show that, in this case, $i = \frac{101}{2} - \frac{5}{d}$. **3**

(b) Hence determine the value(s) of d for which $b_i = 10 + a_i$ is true. **2**

Question 72 (3 marks) DIFFICULT

Adapted from Manly Selective Campus 2026 Trial HSC

The limiting sum of a geometric series is 500, and the sum of the first three terms is 244. Find the sum of the first 10 terms, correct to two decimal places. **3**

Question 73 (3 marks) DIFFICULT

Adapted from Normanhurst Boys High School 2025 Trial HSC

If $y^2 = (2m - y)(2n - y)$, show that $\frac{1}{m}$, $\frac{1}{y}$, $\frac{1}{n}$ (in this order) form the terms of an arithmetic sequence. **3**

Question 74 (3 marks) DIFFICULT

Adapted from North Sydney Boys High School 2025 Trial HSC

Three numbers form a geometric sequence. If the first two terms remain unchanged and the third term is decreased by 32, they form an arithmetic sequence. Then, if the second term of this arithmetic sequence is decreased by 4, the sequence becomes geometric again. Find the third term of the original geometric sequence. **3**

Question 75 (4 marks) DIFFICULT

Adapted from North Sydney Girls High School 2025 Trial HSC

The numbers $4e^{-k},\ 5,\ 4e^{k}$ are consecutive terms in an arithmetic sequence.

(a) Show that $2e^{k}-5+2e^{-k}=0$. **1**

(b) Hence find in exact form any possible values of k. **3**

Question 76 (4 marks) DIFFICULT

Adapted from Pymble Ladies' College 2025 Trial HSC

The first term of an arithmetic series is -20 and the common difference is 5.

(a) Find the sum of the first 20 terms of the progression. **1**

(b) It is given that the sum of the first $2k$ terms is ten times the sum of the first k terms. Find the value of k. **3**

Question 77 (4 marks) DIFFICULT

Adapted from North Sydney Girls High School 2026 Trial HSC

A gambler makes a profit each day but a loss each night. He reduces the amount he bets by a fixed percentage each session, so his profit/loss per session follows a geometric sequence: if his profit is x dollars on the first day, y dollars on the first night and z dollars on the second day, then $x+y+z+\cdots$ is a geometric series.

(a) Given the profit on the third day was \$27 and the loss on the fourth night was \$8, show that the ratio between successive sessions is $-\frac{2}{3}$ and find the profit on the first day. **2**

(b) How much profit or loss will the gambler eventually make? **2**

Question 78 (4 marks) **DIFFICULT**

Adapted from St George Girls High School 2025 Trial HSC

Consider the infinite series $S = \sin^2 x\cos^2 x + 2\sin^2 x\cos^4 x + 4\sin^2 x\cos^6 x + \cdots$

(a) Given that the limiting sum of S exists, show that it is $\dfrac{\sin^2 x\cos^2 x}{\sin^2 x - \cos^2 x}$. **2**

(b) Hence show that $\dfrac{1}{\operatorname{cosec}^2 x(\tan^2 x - 1)}$ can also be expressed as the limiting sum of S. **2**

Question 79 (4 marks) **DIFFICULT**

Adapted from Sydney Grammar School 2026 Trial HSC

Consider the geometric series $\cos x + \cos^3 x + \cos^5 x + \cos^7 x + \cdots$, where $0 < x < \frac{\pi}{2}$.

(a) Show that the limiting sum of this series exists. **1**

(b) Hence solve $\cos x + \cos^3 x + \cos^5 x + \cos^7 x + \cdots = \frac{2}{3}$ for $0 < x < \frac{\pi}{2}$. **3**

End of paper